

Five star tests suffice for the hexagon: the minimum and all attaining interval families

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Abstract

For a convex cyclically labelled hexagon, consider chord multisets of total multiplicity at most three. A multiset is local precisely when its chords triangulate the hexagon. A consecutive interval of two or three vertices detects a multiset if its crossing support chords, together with the two boundary edges of the cut, form a disjoint union of stars. We prove that the minimum size of a fixed interval family detecting every nonlocal multiset is five. We also classify every minimum family: after identifying complementary intervals there are nine, in dihedral orbits of sizes six and three, giving thirty families of the twelve original intervals. Two witness types prove the lower bound; explicit cut tests give elementary upper bounds and the full classification.

1 The covering problem and its scope

Label the vertices of a convex hexagon cyclically by $1, \dots, 6$, with indices interpreted modulo six. Write ij for the unordered chord with endpoints i, j . Only internal diagonals are called chords; thus their set is

$$\mathcal{D} = \{13, 14, 15, 24, 25, 26, 35, 36, 46\}.$$

Let M be a multiset on $\mathcal{D}$ of total multiplicity at most three, and put $S = \text{supp } M$. We call M *local* if it consists of three distinct, pairwise noncrossing chords that triangulate the hexagon, and *nonlocal* otherwise. Crossings here are intersections in the relative interiors of the chords.

For a cyclically consecutive interval T with $2 \leq |T| \leq 3$, form the simple bipartite graph $\Gamma_T(S)$ on the six vertices. Its edges are the chords of S with exactly one endpoint in T , together with the two polygon edges joining T to T^c . The interval *detects* S , or M , when every connected component of this graph is a star. Single edges and isolated vertices are allowed as stars. Define

$$\tau_6 = \min\{|\mathcal{F}| : \text{every nonlocal } M \text{ is detected by some } T \in \mathcal{F}\}, \quad (1)$$

where $\mathcal{F}$ is a family of the twelve allowed intervals. In particular, $\mathcal{F}$ is chosen before the universally quantified multiset.

Rodina, Sellaroli and Valenzuela [1] formulate locality using bounded chord multisets in their equations (6)–(9), introduce this star test in (15), and prove that every nontriangulating support within the chord bound has a detecting interval in Appendix 2, equation (2.1). The paragraph after (9) asks about the minimum number of zeros. We determine the fixed-family star-test covering number at six vertices and all its minimizers. This is a finite refinement of their per-support existence theorem. Our proof uses only the graph definition above. It does not determine the minimum number of physical zero equations if indirect relations among coefficients are also permitted, or an optimum for general polygons.

For notation, set

$$A_i = \{i, i+1\} \quad (1 \leq i \leq 6), \quad B_j = \{j, j+1, j+2\} \quad (1 \leq j \leq 6).$$

The graphs for T and T^c have exactly the same edge set. We therefore identify B_{j+3} with B_j and work first with the nine *quotient cuts* $A_1, \dots, A_6, B_1, B_2, B_3$. Subscripts on quotient B cuts are read modulo three. The minimum in this quotient equals (1): projecting a family cannot increase its size, and choosing one original representative for each quotient cut preserves detection. No minimum family contains both representatives of a complementary pair.

Theorem 1. *The covering number is $\tau_6 = 5$. The minimum quotient families are exactly*

1. *either $\{A_1, A_3, A_5\}$ or $\{A_2, A_4, A_6\}$, together with any two distinct cuts from $\{B_1, B_2, B_3\}$;*
2. *the three rotations of $\{A_1, A_2, A_4, A_5, B_1\}$.*

These types form dihedral orbits of sizes six and three, respectively. Restoring the two complementary representatives of each selected B cut gives exactly thirty minimum families of the original twelve intervals.

For example, the five original intervals $\{1, 2\}, \{2, 3\}, \{4, 5\}, \{5, 6\}, \{1, 2, 3\}$ attain the minimum.

2 Support reduction and explicit cut tests

Detection depends only on S . When $|S| = 3$, the multiplicity bound forces all multiplicities to be one. Three noncrossing chords divide the hexagon into four regions, whose boundary lengths sum to $6 + 2 \cdot 3 = 12$. Each region has at least three sides, so all four are triangles. Thus the domain reduces exactly to the supports $S \subseteq \mathcal{D}$ with $|S| \leq 3$, excluding noncrossing triples. Every smaller support is nonlocal; this includes the empty support and all supports of repeated-chord multisets.

A simple graph is a disjoint union of stars if and only if every edge has an endpoint of degree one. Necessity is immediate. Conversely, every neighbor of a vertex of degree at least two must have degree one, so these vertices constitute a star component; any remaining component is an isolated vertex or a single edge.

Lemma 1. *For every support $S \subseteq \mathcal{D}$, without a cardinality restriction,*

$$A_1 \text{ fails to detect } S \iff 13 \in S \text{ or } 26 \in S \text{ or } \{14, 24\} \subseteq S \text{ or } \{15, 25\} \subseteq S. \quad (2)$$

If $Q = S \cap \{15, 24, 25, 26, 35\}$, then

$$B_1 \text{ detects } S \iff 14, 36 \notin S \text{ and } [Q \subseteq \{15, 24\} \text{ or } Q \subseteq \{26, 35\} \text{ or } Q \subseteq \{25\}]. \quad (3)$$

Rotation gives the tests for all the other cuts.

Proof. The boundary edges of A_1 are 16 and 23. Either 13 or 26 joins them into a four-vertex path, which cannot occur in a star. In their absence, only vertices 1, 2 can have degree greater than one. The graph fails precisely when these possible centers have a common neighbor 4 or 5, giving the two forbidden pairs in (2).

The boundary edges of B_1 are 16 and 34. Chords 13, 46 do not cross the cut. Either 14 or 36 creates a four-vertex path with the boundary edges. Once these two chords are excluded, every singleton in Q is allowed. Among the ten pairs from $\{15, 24, 25, 26, 35\}$, only $\{15, 24\}$ and $\{26, 35\}$ leave star components: in each other pair an edge has both endpoints of degree at least two. No triple has all its pairs among these two disjoint allowed pairs. This proves (3). $\square$

3 Obstructions and the lower bound

Write $D(S)$ for the set of quotient cuts detecting S . Consider

$$P = \{14, 15, 35\}, \quad R = \{15, 24, 46\}.$$

Both are nonlocal: 14 crosses 35 in P , and 15 crosses 46 in R . Lemma 1 and its rotations give exactly

$$D(P) = \{A_1, A_2\}, \quad D(R) = \{A_1, B_1, B_2\}. \quad (4)$$

Rotating P forces every cover $\mathcal{F}$ to satisfy

$$\mathcal{F} \cap \{A_i, A_{i+1}\} \neq \emptyset \quad (1 \leq i \leq 6). \quad (5)$$

Consequently its A cuts form a vertex cover of a six-cycle and number at least three. Equality requires alternation: three vertices supply six incident-edge occurrences, so covering all six edges leaves no edge with both endpoints selected.

The dihedral images of R impose the following twelve conditions: for each row and each listed a , the cover must meet $\{A_a\}$ together with the two indicated B cuts.

B cuts	Possible a
B_1, B_2	1, 3, 4, 6
B_1, B_3	2, 3, 5, 6
B_2, B_3	1, 2, 4, 5

To verify the list, the rotation $v \mapsto v + 1$ sends $A_i \mapsto A_{i+1}$ and $B_j \mapsto B_{j+1}$, whereas the reflection $v \mapsto 2 - v$ sends $A_i \mapsto A_{1-i}$ and $B_j \mapsto B_{-j}$. Apply these to the second detector set in (4). Each entry is thus witnessed by a nonlocal support; no symmetry of $\mathcal{F}$ is assumed.

If no B cut is chosen, the table forces all six A cuts. If only B_1 is chosen, its last row forces A_1, A_2, A_4, A_5 , giving at least five cuts in total; rotation handles either other choice of a single B cut. If at least two B cuts are chosen, (5) again forces at least five cuts. Thus $\tau_6 \geq 5$.

At size five, zero or three B cuts are impossible. Two B cuts require an alternating triple of A cuts; one B cut requires the four A cuts just described. These are exactly the nine families in Theorem 1. We next prove that each really covers the domain.

4 The two upper bounds

4.1 Three alternating A cuts

By rotation take A_1, A_3, A_5 . Put

$$s_1 = 13, \ s_2 = 24, \ s_3 = 35, \ s_4 = 46, \ s_5 = 15, \ s_6 = 26, \quad d_1 = 14, \ d_2 = 25, \ d_3 = 36.$$

We call the d_i diameters, meaning chords between opposite vertices; no regularity assumption is needed. In the next display, each chord name means membership in S . By (2), evasion of all three cuts is equivalent to all three conditions

$$\begin{aligned} A_1 : & \ s_1 \vee s_6 \vee (d_1 \wedge s_2) \vee (s_5 \wedge d_2), \\ A_3 : & \ s_3 \vee s_2 \vee (d_3 \wedge s_4) \vee (s_1 \wedge d_1), \\ A_5 : & \ s_5 \vee s_4 \vee (d_2 \wedge s_6) \vee (s_3 \wedge d_3). \end{aligned} \quad (6)$$

Suppose throughout that $|S| \leq 3$. With at least two diameters there is at most one short chord. Each short chord occurs in only two lines of (6), so simultaneous evasion is impossible.

With exactly one diameter, an even rotation makes it 14 and preserves $\{A_1, A_3, A_5\}$. The last line requires 46 or 15. With one chord slot left, the first two lines require 13 or 24. Hence

$$S = \{14, x, y\}, \quad x \in \{13, 24\}, \quad y \in \{46, 15\}.$$

All four choices are noncrossing triples, so are local.

Without a diameter, the support must contain exactly one chord from each of $\{26, 13\}$, $\{24, 35\}$ and $\{46, 15\}$. Of the eight choices, $\{13, 35, 15\}$ and $\{26, 24, 46\}$ triangulate. Applying (3) to the other six gives the complete list

Supports S	Detecting B cuts
$\{13, 15, 24\}, \{13, 24, 46\}$	B_1, B_2
$\{13, 35, 46\}, \{26, 35, 46\}$	B_1, B_3
$\{15, 24, 26\}, \{15, 26, 35\}$	B_2, B_3

Any two B cuts meet every row. Thus either alternating A triple, together with any two B cuts, detects every nonlocal support. In particular, $\tau_6 \leq 5$.

4.2 Four A cuts and one B cut

Take $\mathcal{F} = \{A_1, A_2, A_4, A_5, B_1\}$. The four A cuts all fail precisely when the following four conditions hold:

$$\begin{aligned}
A_1 : & 13 \vee 26 \vee (14 \wedge 24) \vee (15 \wedge 25), \\
A_2 : & 13 \vee 24 \vee (25 \wedge 35) \vee (26 \wedge 36), \\
A_4 : & 46 \vee 35 \vee (14 \wedge 15) \vee (24 \wedge 25), \\
A_5 : & 46 \vee 15 \vee (25 \wedge 26) \vee (35 \wedge 36).
\end{aligned} \tag{7}$$

We show that a support of size at most three satisfying these conditions is local or contains both 13 and 46.

If $13 \in S$ and $46 \notin S$, the last two lines must be met with at most two further chords. The only completions are $\{15, 35\}$, $\{35, 36\}$ and $\{14, 15\}$; each gives a noncrossing triple with 13. A half-turn gives the case $46 \in S$, $13 \notin S$.

Suppose neither 13 nor 46 is present. With no diameter, (7) forces the four distinct short chords 26, 24, 35, 15, which is impossible. With at least two diameters there is at most one short chord; each of 15, 24, 26, 35 occurs in at most three of the four lines, again impossible. With exactly one diameter the completions are

Diameter	Required short chords
14	$\{15, 24\}$
36	$\{26, 35\}$
25	one from $\{15, 26\}$ and one from $\{24, 35\}$

The six resulting triples are all noncrossing.

Finally, if $\{13, 46\} \subseteq S$, both chords are internal to the sides of B_1 . At most one further chord remains. Formula (3) therefore detects S unless the further chord is 14 or 36; both exceptional triples triangulate. This also covers the two-element support $\{13, 46\}$. We have proved coverage for this family, and rotation gives the other two families of the second type.

4.3 Orbit sizes and complementary representatives

A one-step rotation cycles the three possible B pairs while switching the parity of the alternating A triple. Hence the six first-type families form one orbit. Rotation generates all three second-type families. The number of B cuts is invariant under the dihedral group, so the two types cannot merge. Their dihedral orbit sizes are six and three.

A first-type family has two B cuts, each allowing two independent complementary representatives among the original intervals, and so gives four original families. A second-type family gives two. Since a minimum family contains no complementary pair, these choices omit and duplicate nothing. The total is

$$6 \cdot 2^2 + 3 \cdot 2 = 30.$$

This completes the proof of Theorem 1.

5 Exact verification and limits

The proof above consists of explicit graph tests and finite cases and does not require a solver. Exact enumeration provides a reproducibility check. There are $\binom{12}{3} = 220$ multisets on nine chords of total multiplicity at most three. The support domain has $\sum_{k=0}^3 \binom{9}{k} = 130$ elements; fourteen triangulate, leaving 116 nonlocal supports and 206 nonlocal multisets. Complete enumeration of all 2^9 quotient families and all 2^{12} original families gives the following minimum counts:

Family domain	Minimum size	Number attaining it
Nine quotient cuts	5	9
Twelve original intervals	5	30

The accompanying standard-library Python program `verify.py` reconstructs each cut graph, checks star components directly, and enumerates these domains and the stated families. It uses cyclic endpoint order to test chord crossing, with no floating-point arithmetic or external solver. Run `python3 verify.py` from the manuscript directory. Earlier exact checks also tested the two cut formulas on all 512 supports and all twelve intervals, and a separate implementation used integer geometric crossings and component centers. These checks supplement the displayed proof.

Finite geometric optimization, symmetry reduction and exact graph certification also occur in the adjacent methodological work of Zeng, Liu, Ratnavelu and Huat [2]. That work is cited as broad context for finite geometric certification, without attributing a hexagon star-test result to it.

The source comparison here concerns the inspected version arXiv:2608.23008v1. Its cited existence theorem and minimum-number question provide the specific context; no priority claim or completed wider literature survey is asserted. The ancillary program associated with that source was not inspected. The theorem resolves the stated six-vertex covering problem; the general polygon problem and possible reductions using indirect physical coefficient relations remain outside its scope.

Computational and AI assistance. OpenAI Codex assisted with proof exploration, exact verification programs, manuscript drafting and local build checks. The mathematical record also underwent a separate model-context review with a separately implemented exact enumeration. Such review is not external peer review, and no proof-assistant formalization is claimed. The elementary argument and finite checks are provided for direct inspection.

References

- [1] L. Rodina, G. Sellaroli and J. Valenzuela, *A Combinatorial Origin of Locality*, arXiv:2608.23008v1 [hep-th] (2026). <https://arxiv.org/abs/2608.23008v1>.
- [2] Zijian Zeng, Houde Liu, Kuru Ratnavelu and Ong Seng Huat, *Exact Finite Certificates for the Four-Dimensional Ternary Borsuk Problem and Five-Dimensional Ternary Kissing Codes*, Preprints (2026). <https://doi.org/10.20944/preprints202608.1658.v1>.